

Assignment 1

Course: CAD for VLSI Design (PVL 113)

Course Instructor: Dr. Sandeep Mandia

1. Describe the architecture of the UNIX operating system with a suitable diagram.
2. What does the "ls" command do? Describe -l and -a options for the same with suitable examples. Explain the file type and permission sequence.
3. Consider the following directory structure. Suppose you are currently in the directory C. Which command will you use to know your current directory, and how will you move to directory F (write three ways). Write the output for each command.

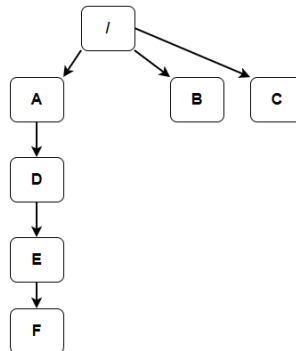


Figure 1

4. Write a sequence of commands to make the directory structure in Figure 1 from the root. How can you create the same directory structure using only three commands?
5. Explain the following commands with examples: wc, cp, mv, rm.
6. Describe the process of creating a file, adding content to it and saving it using the vi editor. Mention different ways to exit from the file.
7. The content of a file with filename "abc.txt" is shown in figure 2. Suppose the cursor is at the character "a" of word and in second line and you execute the following commands in the vi editor. Write content of the file after each command.
(i) 2d (ii) d\$ (iii) D (iv) dd (v) 5l, x

CAD for VLSI course is taught to 1st year M. Tech
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Figure 2

8. Write a UNIX shell script that takes a year as input and checks whether it's a leap. Display appropriate messages based on the input.
9. Write a shell script that takes three numbers as inputs and displays the largest/smallest of them.
10. Write a shell script to implement the problem 9 using functions.
11. Write a shell script to compute the factorial of a number using function recursion.
12. Write a shell script to print the following pattern. Use for loop.

```
0
1 0
2 1 0
3 2 1 0
```

4 3 2 1 0

5 4 3 2 1 0

13. Write a shell script that uses a `while` loop to calculate the sum of all even numbers between 1 and 100. The script should display the final sum at the end.
14. Write a shell script that uses a select loop to present a menu to the user with the following options:
 - a. Display the current date and time.
 - b. List all files in the current directory.
 - c. Show the current working directory.
 - d. Exit the script.

The script should perform the selected action and continue displaying the menu until the user chooses to exit.

15. Write `sed` commands that perform the following tasks:
 - a. Replace all occurrences of the word "Linux" with "UNIX" in a given file `sample.txt`.
 - b. command to delete all blank lines from a file `sample.txt`.
 - c. To print lines from 5 to 10 in a file `logfile.txt`.
16. Write `grep` commands that perform the following tasks:
 - a. To find all lines in the file `sample.txt` that contain the word "error".
 - b. To search for the word "hello" in a file `messages.txt`, ignoring case differences.
 - c. To display only the lines that start with the word "ERROR" in the file `log.txt`.